```
import os
os.environ["GEMINI_API_KEY"] = "AQ.Ab8RN6JQSHOGdjMbYiSOrjsQcqT8pdrz_efFgV8ljaehELHz-Q"
```

```
!pip install -q \
youtube-transcript-api \
langchain-community \
langchain-text-splitters\
langchain-google-genai \
google-generativeai \
faiss-cpu \
tiktoken \
python-dotenv
```

```
from youtube_transcript_api import YouTubeTranscriptApi, TranscriptsDisabled
from langchain_text_splitters import RecursiveCharacterTextSplitter
from langchain_google_genai import GoogleGenerativeAIEmbeddings, ChatGoogleGenerativeAI
from langchain_community.vectorstores import FAISS
from langchain_core.prompts import PromptTemplate
```

## Step 1a - Indexing (Document Ingestion)

```
video_id = "Gfr50f6ZBvo"

ytt_api = YouTubeTranscriptApi()

transcript = ytt_api.fetch(video_id)

text = " ".join(snippet.text for snippet in transcript)

print(text)
```

```
the following is a conversation with demus hasabis ceo and co-founder of deepmind a company that has published and builds some of the most incredible artificial intelli
```

```
print(transcript)
```

```
FetchedTranscript(snippets=[FetchedTranscriptSnippet(text='the following is a conversation with', start=0.08, duration=3.44), FetchedTranscriptSnippet(text='demus hasab
```

## Step 1b - Indexing (Text Splitting)

```
splitter = RecursiveCharacterTextSplitter(
    chunk_size=3000
```

```
    chunk_overlap=300
)

chunks = splitter.create_documents([text])

print(len(chunks))
```

```
50
```

```
chunks[49]
```

```
Document(metadata={}, page_content="more fundamental explanations of physics would be the beginning you know more careful specification of that taking you walking us through by the hand as to what one would do to maybe prove those things out maybe giving you glimpses of what things you totally missed in the physics of today exactly just here here's glimpses of no like there's a much uh a much more elaborate world or a much simpler world or something a much deeper maybe simpler explanation yes of things right than the standard model of physics which we know doesn't work but we still keep adding to so um and and that's how i think the beginning of an explanation would look and it would start encompassing many of the mysteries that we have wondered about for thousands of years like you know consciousness uh life and gravity all of these things yeah giving us a glimpses of explanations for those things yeah well um damas dear one of the special human beings in this giant puzzle of ours and it's a huge honor that you would take a pause from the bigger puzzle to solve this small puzzle of a conversation with me today it's truly an honor and a pleasure thank you thank you i really enjoyed it thanks lex thanks for listening to this conversation with demas establish to support this podcast please check out our sponsors in the description and now let me leave you with some words from edskar dykstra computer science is no more about computers than astronomy is about telescopes thank you for listening and hope to see you next time")
```

## Step 1c & 1d - Indexing (Embedding Generation and Storing in Vector Store)

```
embeddings = GoogleGenerativeAIEmbeddings(model= "models/gemini-embedding-001")
vector_store = FAISS.from_documents(chunks, embeddings)
```

```
---------------------------------------------------------------------------
ClientError                               Traceback (most recent call last)
/usr/local/lib/python3.12/dist-packages/langchain_google_genai/embeddings.py in embed_documents(self, texts, batch_size, task_type, titles, output_dimensionality)
    448             try:
--> 449                 result = self.client.models.embed_content(
    450                     model=self.model,

17 frames

ClientError: 429 RESOURCE_EXHAUSTED. {'error': {'code': 429, 'message': 'You exceeded your current quota, please check your plan and billing details. For more information on this error, head to: https://ai.google.dev/gemini-api/docs/rate-limits. To monitor your current usage, head to: https://ai.dev/rate-limit. ', 'status': 'RESOURCE_EXHAUSTED', 'details': [{'@type': 'type.googleapis.com/google.rpc.Help', 'links': [{'description': 'Learn more about Gemini API quotas', 'url': 'https://ai.google.dev/gemini-api/docs/rate-limits'}]}]}}

The above exception was the direct cause of the following exception:

GoogleGenerativeAIError                   Traceback (most recent call last)
/usr/local/lib/python3.12/dist-packages/langchain_google_genai/embeddings.py in embed_documents(self, texts, batch_size, task_type, titles, output_dimensionality)
    454             except ClientError as e:
    455                 msg = f"Error embedding content ({e.status}): {e}"
--> 456                 raise GoogleGenerativeAIError(msg) from e
    457             except Exception as e:
    458                 msg = f"Error embedding content: {e}"

GoogleGenerativeAIError: Error embedding content (RESOURCE_EXHAUSTED): 429 RESOURCE_EXHAUSTED. {'error': {'code': 429, 'message': 'You exceeded your current quota, please check your plan and billing details. For more information on this error, head to: https://ai.google.dev/gemini-api/docs/rate-limits. To monitor your current usage, head to: https://ai.dev/rate-limit. ', 'status': 'RESOURCE_EXHAUSTED', 'details': [{'@type': 'type.googleapis.com/google.rpc.Help', 'links': [{'description': 'Learn more about Gemini API quotas', 'url': 'https://ai.google.dev/gemini-api/docs/rate-limits'}]}]}}
```

Next steps: Explain error

```
print(len(chunks))
```

```
50
```

```
vector_store.index_to_docstore_id
```

```
{0: 'f5bb5947-b586-404b-97c2-1bab067621d0',
 1: 'a7e1c485-d6d2-4fef-944f-5ce9e89c9db8',
 2: '2495f945-f964-49a8-97bd-39f1c659b126',
 3: 'c5bc9278-ed5f-4cb5-9573-f747205dd931',
 4: 'cb395694-7c2e-42d1-a437-b5d8f1ee5110',
 5: '12bee84b-bde3-421b-9a5c-2310c0470ca0',
 6: '36010522-d64d-4f11-b910-609edc79503f',
 7: '7a8cdd59-005a-4f6a-946d-88a3ab4a3b9f',
 8: '9b85be3e-ffeb-45e4-a3cf-df7dc8509b42',
 9: '8b8869b4-578a-4699-bafe-4bd4a067cc8d',
 10: '887c4e02-e55b-412b-9895-773524a27cdd',
 11: 'c916ad64-7d99-4604-8ca5-106b41229b12',
 12: '7c832feb-935f-44ba-8075-57641470842f',
 13: 'e195d7e8-e8e0-4d0d-8c1f-39b6a8c816c3',
 14: '2b9612af-e457-40c5-8794-9f2a43f799e9',
```

```
 15: '327a147f-8844-4a64-acbb-f8f1856ac15e',
 16: 'c2de65f1-9337-4586-b1b8-5900f806ef36',
 17: 'e78493fd-2eae-41c0-a822-96c30b4a2998',
 18: '41258426-96de-4065-a28b-973a95d60f4d',
 19: '7c77bd03-60bb-48a9-9825-026844f3516f',
 20: '3a33eb2b-411d-4c6a-b11f-5eb7f731efc4',
 21: '8d886c80-bee2-4327-b05e-dfe9558969af',
 22: '9cd1637a-e821-4b38-97ca-217bd6bd664c',
 23: '154cf8f4-9f70-462d-a77a-cb1e7c61abe1',
 24: 'bac09ca7-b7b0-4006-8066-1b719d171b4c',
 25: '439fbc55-10cc-426b-a809-39b75868faf3',
 26: 'fcc1b3fd-8974-4b36-a128-c90f096c0c45',
 27: 'e80c970f-ebbf-4726-9fbf-bcc04868b6ff',
 28: '8ee9a4cd-e430-4b55-8235-0be64e56acbb',
 29: '95ebe010-42be-4991-8001-42d8fbd4acac',
 30: 'ebc501fe-67b6-4d16-a825-a702de604b53',
 31: 'a1cc2b83-ab8e-4bf5-860f-a3d04db27488',
 32: '68a16014-3915-49c2-923d-68ae5d62ae7d',
 33: 'd403894a-0888-47fa-8d57-58d62775cafc',
 34: '297da8e2-9808-4696-8d85-9a5de1cae79e',
 35: 'f77a9b4c-edb4-4a7f-bc0f-e43b50f375eb',
 36: '0be67484-85e1-47ad-8893-c7908751e1ee',
 37: 'd4773958-55db-42eb-acc1-b6c64e859ac9',
 38: 'eedbf569-1cdd-46ae-9e74-de0c43c317a3',
 39: 'cef48e93-3749-41ad-b9f8-854a6f192c15',
 40: '050277e7-2182-49d0-960a-90894559691f',
 41: '4421aed1-93ea-4071-8815-7942d4cb20cd',
 42: '32739d65-59fd-43f4-a276-7880f8bb3599',
 43: '62a490a3-80d4-4841-9124-df03d4b59aff',
 44: '7c9bb43d-2b10-40e8-ab75-321bcc09d291',
 45: '22d980d9-c321-4fc7-9192-68f63154589f',
 46: 'f691a5f1-8cbf-43cb-be9a-db4e94e41051',
 47: '039a2f5e-7412-4d91-8256-b61633826729',
 48: 'd4bed8e8-5cc0-40cf-ac95-1d803d062aab',
 49: 'e98cb8e0-efe3-4a25-b7e5-7a2cd83ec5ae'}
```

```python
vector_store.get_by_ids(['e98cb8e0-efe3-4a25-b7e5-7a2cd83ec5ae'])
```

```
[Document(id='e98cb8e0-efe3-4a25-b7e5-7a2cd83ec5ae', metadata={}, page_content="more fundamental explanations of physics would be the beginning you know more careful
specification of that taking you walking us through by the hand as to what one would do to maybe prove those things out maybe giving you glimpses of what things you
totally missed in the physics of today exactly just here here's glimpses of no like there's a much uh a much more elaborate world or a much simpler world or something
a much deeper maybe simpler explanation yes of things right than the standard model of physics which we know doesn't work but we still keep adding to so um and and
that's how i think the beginning of an explanation would look and it would start encompassing many of the mysteries that we have wondered about for thousands of years
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with me today it's truly an honor and a pleasure thank you thank you i really enjoyed it thanks lex thanks for listening to this conversation with demas establish to
support this podcast please check out our sponsors in the description and now let me leave you with some words from edskar dykstra computer science is no more about
computers than astronomy is about telescopes thank you for listening and hope to see you next time")]
```

## Step 2 - Retrieval

```python
retriever = vector_store.as_retriever(search_type="similarity", search_kwargs={"k": 4})
```

```python
retriever
```

```
VectorStoreRetriever(tags=['FAISS', 'GoogleGenerativeAIEmbeddings'], vectorstore=<langchain_community.vectorstores.faiss.FAISS object at 0x7c9c85855580>,
search_kwargs={'k': 4})
```

```python
retriever.invoke('What is deepmind')
```

```
several you know a decade ago i remember someone claiming that with a with a kind of very bog standard normal uh"),
 Document(id='7c77bd03-60bb-48a9-9825-026844f3516f', metadata={}, page_content="to think to this because you're in a very interesting position where deepmind is the
place of some of the most uh brilliant ideas in the history of ai but it's also a place of brilliant engineering so how much of solving intelligence this big goal for
deepmind how much of it is science how much is engineering so how much is the algorithms how much is the data how much is the hardware compute infrastructure how much
is it the software computer infrastructure yeah um what else is there how much is the human infrastructure and like just the humans interact in certain kinds of ways
in all the space of all those ideas how much does maybe like philosophy how much what's the key if um uh if if you were to sort of look back like if we go forward 200
years look back what was the key thing that solved intelligence is that ideas i think it's a combination first of all of course it's a combination of all those things
but the the ratios of them changed over over time so yeah so um even in the last 12 years so we started deep mine in 2010 which is hard to imagine now because 2010
it's only 12 short years ago but nobody was talking about ai uh you know if you remember back to your mit days you know no one was talking about it i did a postdoc at
mit back around then and it was sort of thought of as a well look we know ai doesn't work we tried this hard in the 90s at places like mit mostly losing using logic
systems and old-fashioned sort of good old-fashioned ai we would call it now um people like minsky and and and patrick winston and you know all these characters right
and used to debate a few of them and they used to think i was mad thinking about that some new advance could be done with learning systems and um i was actually
pleased to hear that because at least you know you're on a unique track at that point right even if every all of your you know professors are telling you you're mad
that's true and of course in industry uh you can we couldn't get you know as difficult to get two cents together uh and which is hard to imagine now as well given
it's the biggest sort of buzzword in in vcs and and fundraising's easy and all these kind of things today so back in 2010 it was very difficult and what we the reason
we started then and shane and i used to discuss um uh uh what were the sort of founding tenets of deep mind and it was very various things one was um algorithmic
advances so deep learning you know jeff hinton and cohen just had just sort of invented that in academia but no one in industry knew about it uh we love reinforcement
learning we thought that could be scaled up but also understanding about the human brain had advanced um quite a lot uh in the decade prior with fmri machines and
other things so we could get some good hints about architectures and algorithms and and sort of um representations maybe that the brain uses so as at a systems level
not at a implementation level um and then the other big things were compute and gpus right so we could"),
 Document(id='2b9612af-e457-40c5-8794-9f2a43f799e9', metadata={}, page_content="i think it's one of the biggest breakthroughs uh certainly in the history of
structural biology but uh in general in in science um maybe from a high level what is it and how does it work and then we can ask some fascinating sure questions
after sure um so maybe like to explain it uh to people not familiar with protein folding is you know i first of all explain proteins which is you know proteins are
essential to all life every function in your body depends on proteins sometimes they're called the workhorses of biology and if you look into them and i've you know
obviously as part of alpha fold i've been researching proteins and and structural biology for the last few years you know they're amazing little bio nano machines
proteins they're incredible if you actually watch little videos of how they work animations of how they work and um proteins are specified by their genetic sequence
called the amino acid sequence so you can think of those their genetic makeup and then in the body uh in in nature they when they when they fold up into a 3d
```

```
limits of what classical computing can do now um and at the same time i've also studied neuroscience to see and that's why i did my phd in was to see also to look at
you know is there anything quantum in the brain from a neuroscience or biological perspective and um and so far i think most neuroscientists and most mainstream
biologists and neuroscientists would say there's no evidence of any quantum uh systems or effects in the brain as far as we can see it's it can be mostly explained by
classical uh classical theories so and then so there's sort of the the search from the biology side and then at the same time there's the raising of the water uh at
the bar from what classical turing machines can do uh uh and and you know including our new ai systems and uh as you alluded to earlier um you know i think ai
especially in the last decade plus has been a continual story now of surprising uh events uh and surprising successes knocking over one theory after another of what
was thought to be impossible you know from go to protein folding and so on and so i think um i would be very hesitant to bet against how far the uh universal turing
machine and classical computation paradigm can go and and my betting would be that all of certainly what's going on in our brain uh can probably be mimicked or or
approximated on a on a classical machine um not you know not requiring something metaphysical or quantum and we'll get there with some of the work with alpha fold
which i think begins the journey of modeling this beautiful and complex world of biology so you think all the magic of the human mind comes from this just a few
pounds of mush of a biological computational mush that's akin to some of the neural networks not directly but in spirit that deep mind has been working with well look
i think it's um you say it's a few you know of course it's this is the i think the biggest miracle of the universe is that um it is just a few pounds of mush in our
skulls and yet it's also our brains are the most complex objects in the in that we know of in the universe so there's")]
```

## Step 3 - Augmentation

```
llm = ChatGoogleGenerativeAI(model="gemini-2.5-flash")
```

```
prompt = PromptTemplate(
    template="""
      You are a helpful assistant.
      Answer ONLY from the provided transcript context.
      If the context is insufficient, just say you don't know.

      {context}
      Question: {question}
    """,
    input_variables = ['context', 'question']
)
```

```
question = "is the topic of nuclear fusion discussed in this video? if yes then what was discussed"
retrieved_docs = retriever.invoke(question)
```

```
retrieved_docs
```

```
simple molecules we would like to simulate large materials um and so uh today there's no way of doing that and we're building up towards uh building functionals that
```

```
simple molecules we would like to simulate large materials um and so uh today there's no way of doing that and we're building up towards uh building functionals that
approximate schrodinger's equation and then allow you to describe uh what the electrons are doing and all materials sort of science and material properties are
governed by the electrons and and how they interact so have a good summarization of the simulation through the functional um but one that is still close to what the
actual simulation would come out with so what um how difficult is that to ask what's involved in that task is it running those those complicated simulations yeah and
learning the task of mapping from the initial conditions and the parameters of the simulation learning what the functional would be yeah so it's pretty tricky and
we've done it with um you know the nice thing is we there are we can run a lot of the simulations that the molecular dynamics"),
 Document(id='e80c970f-ebbf-4726-9fbf-bcc04868b6ff', metadata={}, page_content="somewhere but just a unique way of putting those things together i think some of the
greatest scientists in history have displayed that i would say although it's very hard to know going back to their time what was exactly known uh when they came up
with those things although you're making me really think because just the thought experiment of deeply knowing a hundred wikipedia pages i don't think i can um i've
been really impressed by wikipedia for for technical topics yeah so if you know a hundred pages or a thousand pages i don't think who can visually truly comprehend
what's what kind of intelligence that is that's a pretty powerful intelligence if you know how to use that and integrate that information correctly yes i think you
can go really far you can probably construct thought experiments based on that like simulate different ideas so if this is true let me run this thought experiment
then maybe this is true it's not really invention it's like just taking literally the knowledge and using it to construct a very basic simulation of the world i mean
some argue it's romantic in part but einstein would do the same kind of things with a thought experiment yeah one could imagine doing that systematically across
millions of wikipedia pages plus pubmed all these things i think there are many many things to be discovered like that they're hugely useful you know you could
imagine and i want us to do some of those things in material science like room temperature superconductors or something on my list one day i'd like to like you know
have an ai system to help build better optimized batteries all of these sort of mechanical things mr i think a systematic sort of search could be uh guided by a model
could be um could be extremely powerful so speaking of which you have a paper on nuclear fusion uh magnetic control of tokamak plasmas to deep reinforcement learning
so you uh you're seeking to solve nuclear fusion with deep rl so it's doing control of high temperature plasmas can you explain this work and uh can ai eventually
solve nuclear fusion it's been very fun last year or two and very productive because we've been taking off a lot of my dream projects if you like of things that i've
collected over the years of areas of science that i would like to i think could be very transformative if we helped accelerate and uh really interesting problems
scientific challenges in of themselves this is energy so energy yes exactly so energy and climate so we talked about disease and biology as being one of the biggest
places i think ai can help with i think energy and climate uh is another one so maybe they would be my top two um and fusion is one one area i think ai can help with
now fusion has many challenges mostly physics material science and engineering challenges as well to build these massive fusion reactors and contain the plasma and
what we try to do whenever we go into a new field to apply our systems is we look for um we talk to domain experts we try"),
 Document(id='f5bb5947-b586-404b-97c2-1bab067621d0', metadata={}, page_content="the following is a conversation with demus hasabis ceo and co-founder of deepmind a
company that has published and builds some of the most incredible artificial intelligence systems in the history of computing including alfred zero that learned all
by itself to play the game of gold better than any human in the world and alpha fold two that solved protein folding both tasks considered nearly impossible for a
very long time demus is widely considered to be one of the most brilliant and impactful humans in the history of artificial intelligence and science and engineering
in general this was truly an honor and a pleasure for me to finally sit down with him for this conversation and i'm sure we will talk many times again in the future
this is the lex friedman podcast to support it please check out our sponsors in the description and now dear friends here's demis hassabis let's start with a bit of a
personal question am i an ai program you wrote to interview people until i get good enough to interview you well i'll be impressed if if you were i'd be impressed by
myself if you were i don't think we're quite up to that yet but uh maybe you're from the future lex if you did would you tell me is that is that a good thing to tell
a language model that's tasked with interviewing that it is in fact um ai maybe we're in a kind of meta turing test uh probably probably it would be a good idea not
to tell you so it doesn't change your behavior right this is a kind of heisenberg uncertainty principle situation if i told you you behave differently yeah maybe
that's what's happening with us of course this is a benchmark from the future where they replay 2022 as a year before ais were good enough yet and now we want to see
is it going to pass exactly if i was such a program would you be able to tell do you think so to the touring test question you've talked about the benchmark for
solving intelligence what would be the impressive thing you've talked about winning a nobel prize in a system winning a nobel prize but i still return to the touring
test as a compelling test the spirit of the touring test is a compelling test yeah the turing test of course it's been unbelievably influential and turing's one of my
all-time heroes but i think if you look back at the 1950 papers original paper and read the original you'll see i don't think he meant it to be a rigorous formal test
i think it was more like a thought experiment almost a bit of philosophy he was writing if you look at the style of the paper and you can see he didn't specify it
very rigorously so for example he didn't specify the knowledge that the expert or judge would have um not you know how much time would they have to investigate this
so these important parameters if you were gonna make it uh a true sort of formal test um and you know some by some measures people claimed the turing test passed
several you know a decade ago i remember someone claiming that with a with a kind of very bog standard normal uh")]
```

```
context_text = "\n\n".join(doc.page_content for doc in retrieved_docs)
context_text
```

```
'can help with now fusion has many challenges mostly physics material science and engineering challenges as well to build these massive fusion reactors and contain the
plasma and what we try to do whenever we go into a new field to apply our systems is we look for um we talk to domain experts we try and find the best people in the wo
rld to collaborate with um in this case in fusion we we collaborated with epfl in switzerland the swiss technical institute who are amazing they have a test reactor th
at they were willing to let us use which you know i double checked with the team we were going to use carefully and safely i was impressed they managed to persuade the
m to let us use it and um and it's a it's an amazing test reactor they have there and they try all sorts of pretty crazy experiments on it and um the the the what we t
end to look at is if we go into a new domain like fusion what are all the bottleneck problems uh like thinking from first principles you know what are all the bottle…
```

```
final_prompt = prompt.invoke({"context": context_text, "question": question})
```

```
final_prompt
```

```
StringPromptValue(text="\n      You are a helpful assistant.\n      Answer ONLY from the provided transcript context.\n      If the context is insufficient, just say
you don't know.\n\n      can help with now fusion has many challenges mostly physics material science and engineering challenges as well to build these massive fusion
reactors and contain the plasma and what we try to do whenever we go into a new field to apply our systems is we look for um we talk to domain experts we try and find
the best people in the world to collaborate with um in this case in fusion we we collaborated with epfl in switzerland the swiss technical institute who are amazing
they have a test reactor that they were willing to let us use which you know i double checked with the team we were going to use carefully and safely i was impressed
they managed to persuade them to let us use it and um and it's a it's an amazing test reactor they have there and they try all sorts of pretty crazy experiments on it
and um the the the what we tend to look at is if we go into a new domain like fusion what are all the bottleneck problems uh like thinking from first principles you
know what are all the bottleneck problems that are still stopping fusion working today and then we look at we you know we get a fusion expert to tell us and then we
look at those bottlenecks and we look at the ones which ones are amenable to our ai methods today yes right and and and then and would be interesting from a research
perspective from our point of view from an ai point of view and that would address one of their bottlenecks and in this case plasma control was was perfect so you
know the plasma it's a million degrees celsius something like that it's hotter than the sun and there's obviously no material that can contain it so they have to be
containing these magnetic very powerful superconducting magnetic fields but the problem is plasma is pretty unstable as you imagine you're kind of holding a mini sun
mini star in a reactor so you know you you kind of want to predict ahead of time what the plasma's going to do so you can move the magnetic field within a few
milliseconds you know to to basically contain what it's going to do next so it seems like a perfect problem if you think of it for like a reinforcement learning
prediction problem so uh you know your controller you're gonna move the magnetic field and until we came along you know they were they were doing it with with
traditional operational uh research type of uh controllers uh which are kind of handcrafted and the problem is of course they can't react in the moment to something
the plasma's doing that they have to be hard-coded and again knowing that that's normally our go-to solution is we would like to learn that instead and they also had
a simulator of these plasma so there were lots of criteria that matched what we we like to to to use so can ai eventually solve nuclear fusion well so we with this
problem and we published it in a nature paper last year uh we held the fusion that we held the plasma in specific shapes so actually it's almost like carving the
plasma into different shapes and control and hold it there for the\n\nto use so can ai eventually solve nuclear fusion well so we with this problem and we published
it in a nature paper last year uh we held the fusion that we held the plasma in specific shapes so actually it's almost like carving the plasma into different shapes
and control and hold it there for the record amount of time so um so that's one of the problems of of fusion sort of um solved so i have a controller that's able to
no matter the shape uh contain it continue yeah contain it and hold it in structure and there's different shapes that are better for for the energy productions called
droplets and and and so on so um so that was huge and now we're looking we're talking to lots of fusion startups to see what's the next problem we can tackle uh in
the fusion area so another fascinating place in a paper title pushing the frontiers of density functionals by solving the fractional electron problem so you're taking
on modeling and simulating the quantum mechanical behavior of electrons yes um can you explain this work and can ai model and simulate arbitrary quantum mechanical
systems in the future yeah so this is another problem i've had my eye on for you know a decade or more which is um uh sort of simulating the properties of electrons
if you can do that you can basically describe how elements and materials and substances work so it's kind of like fundamental if you want to advance material science
um and uh you know we have schrodinger's equation and then we have approximations to that density functional theory these things are you know are famous and um people
try and write approximations to to these uh uh to these functionals and and kind of come up with descriptions of the electron clouds where they're gonna go how
they're gonna interact when you put two elements together uh and what we try to do is learn a simulation uh uh learner functional that will describe more chemistry
types of chemistry so um until now you know you can run expensive simulations but then you can only simulate very small uh molecules very simple molecules we would
like to simulate large materials um and so uh today there's no way of doing that and we're building up towards uh building functionals that approximate schrodinger's
equation and then allow you to describe uh what the electrons are doing and all materials sort of science and material properties are governed by the electrons and
and how they interact so have a good summarization of the simulation through the functional um but one that is still close to what the actual simulation would come
out with so what um how difficult is that to ask what's involved in that task is it running those those complicated simulations yeah and learning the task of mapping
from the initial conditions and the parameters of the simulation learning what the functional would be yeah so it's pretty tricky and we've done it with um you know
the nice thing is we there are we can run a lot of the simulations that the molecular dynamics\n\nsomewhere but just a unique way of putting those things together i
think some of the greatest scientists in history have displayed that i would say although it's very hard to know going back to their time what was exactly known uh
when they came up with those things although you're making me really think because just the thought experiment of deeply knowing a hundred wikipedia pages i don't
think i can um i've been really impressed by wikipedia for for technical topics yeah so if you know a hundred pages or a thousand pages i don't think who can visually
truly comprehend what's what kind of intelligence that is that's a pretty powerful intelligence if you know how to use that and integrate that information correctly
yes i think you can go really far you can probably construct thought experiments based on that like simulate different ideas so if this is true let me run this
thought experiment then maybe this is true it's not really invention it's like just taking literally the knowledge and using it to construct a very basic simulation
of the world i mean some argue it's romantic in part but einstein would do the same kind of things with a thought experiment yeah one could imagine doing that
systematically across millions of wikipedia pages plus pubmed all these things i think there are many many things to be discovered like that they're hugely useful you
know you could imagine and i want us to do some of those things in material science like room temperature superconductors or something on my list one day i'd like to
like you know have an ai system to help build better optimized batteries all of these sort of mechanical things mr i think a systematic sort of search could be uh
guided by a model could be um could be extremely powerful so speaking of which you have a paper on nuclear fusion uh magnetic control of tokamak plasmas to deep
reinforcement learning so you uh you're seeking to solve nuclear fusion with deep rl so it's doing control of high temperature plasmas can you explain this work and
uh can ai eventually solve nuclear fusion it's been very fun last year or two and very productive because we've been taking off a lot of my dream projects if you like
```

```
of things that i've collected over the years of areas of science that i would like to i think could be very transformative if we helped accelerate and uh really
interesting problems scientific challenges in of themselves this is energy so energy yes exactly so energy and climate so we talked about disease and biology as being
one of the biggest places i think ai can help with i think energy and climate uh is another one so maybe they would be my top two um and fusion is one one area i
think ai can help with now fusion has many challenges mostly physics material science and engineering challenges as well to build these massive fusion reactors and
contain the plasma and what we try to do whenever we go into a new field to apply our systems is we look for um we talk to domain experts we try\n\nthe following is a
conversation with demus hasabis ceo and co-founder of deepmind a company that has published and builds some of the most incredible artificial intelligence systems in
the history of computing including alfred zero that learned all by itself to play the game of gold better than any human in the world and alpha fold two that solved
```

## Step 4 - Generation

```
answer = llm.invoke(final_prompt)
print(answer.content)
```

```
Yes, the topic of nuclear fusion is discussed in the video.

The discussion covers:
*   **Challenges of Fusion:** It's highlighted that fusion faces significant challenges in physics, material science, and engineering, especially in building large reac
*   **AI's Role:** DeepMind views fusion as a transformative area where AI can significantly help, placing it among the top two areas (alongside disease and biology) wh
*   **Collaboration and Approach:** DeepMind collaborated with EPFL in Switzerland, using their test reactor. They focus on identifying bottleneck problems in fusion th
*   **Plasma Control:** The primary bottleneck problem identified and tackled was plasma control. Plasma is described as being extremely hot (millions of degrees Celsiu
*   **Deep Reinforcement Learning (RL):** This problem was deemed perfect for reinforcement learning. Traditional methods used "handcrafted" controllers that couldn't r
*   **DeepMind's Achievement:** DeepMind published a Nature paper detailing their work using deep reinforcement learning for magnetic control of tokamak plasmas. They s
*   **Future Work:** They are currently exploring the next problems to tackle in the fusion area by engaging with fusion startups.
```

## Building a Chain

```
from langchain_core.runnables import RunnableParallel, RunnablePassthrough, RunnableLambda
from langchain_core.output_parsers import StrOutputParser
```

```
def format_docs(retrieved_docs):
  context_text = "\n\n".join(doc.page_content for doc in retrieved_docs)
  return context_text
```

```
parallel_chain = RunnableParallel({
    'context': retriever | RunnableLambda(format_docs),
    'question': RunnablePassthrough()
})
```

```
parallel_chain.invoke('who is Demis')
```

```
{'context': "the following is a conversation with demus hasabis ceo and co-founder of deepmind a company that has published and builds some of the most incredible
artificial intelligence systems in the history of computing including alfred zero that learned all by itself to play the game of gold better than any human in the
world and alpha fold two that solved protein folding both tasks considered nearly impossible for a very long time demus is widely considered to be one of the most
brilliant and impactful humans in the history of artificial intelligence and science and engineering in general this was truly an honor and a pleasure for me to
finally sit down with him for this conversation and i'm sure we will talk many times again in the future this is the lex friedman podcast to support it please check
```

```
out our sponsors in the description and now dear friends here's demis hassabis let's start with a bit of a personal question am i an ai program you wrote to interview
people until i get good enough to interview you well i'll be impressed if if you were i'd be impressed by myself if you were i don't think we're quite up to that yet
but uh maybe you're from the future lex if you did would you tell me is that is that a good thing to tell a language model that's tasked with interviewing that it is
in fact um ai maybe we're in a kind of meta turing test uh probably probably it would be a good idea not to tell you so it doesn't change your behavior right this is
a kind of heisenberg uncertainty principle situation if i told you you behave differently yeah maybe that's what's happening with us of course this is a benchmark
from the future where they replay 2022 as a year before ais were good enough yet and now we want to see is it going to pass exactly if i was such a program would you
be able to tell do you think so to the touring test question you've talked about the benchmark for solving intelligence what would be the impressive thing you've
talked about winning a nobel prize in a system winning a nobel prize but i still return to the touring test as a compelling test the spirit of the touring test is a
compelling test yeah the turing test of course it's been unbelievably influential and turing's one of my all-time heroes but i think if you look back at the 1950
papers original paper and read the original you'll see i don't think he meant it to be a rigorous formal test i think it was more like a thought experiment almost a
bit of philosophy he was writing if you look at the style of the paper and you can see he didn't specify it very rigorously so for example he didn't specify the
knowledge that the expert or judge would have um not you know how much time would they have to investigate this so these important parameters if you were gonna make
it uh a true sort of formal test um and you know some by some measures people claimed the turing test passed several you know a decade ago i remember someone claiming
that with a with a kind of very bog standard normal uh\n\nmade of the same things biological neurons so we're gonna have to come up with explanations uh or models of
the gap between substrate differences between machines and humans did to get anywhere beyond the behavioral but to me sort of the practical question is very
interesting and very important when you have millions perhaps billions of people believing that you have ascension ai believing what that google engineer believed
which i just see is an obvious very near-term future thing certainly on the path to agi how does that change the world what's the responsibility of the ai system to
help those millions of people and also what's the ethical thing because you can you can make a lot of people happy by creating a meaningful deep experience with a
system that's faking it before it makes it yeah and i i don't is a are we the right or who is to say what's the right thing to do should ai always be tools like why
why why are we constraining ais to always be tools as opposed to friends yeah i think well i mean these are you know you know fantastic questions and and also
critical ones and we've been thinking about this uh since the start of d minor before that because we planned for success and you know how how you know you know
however remote that looked like back in 2010 and we've always had sort of these ethical considerations as fundamental at deepmind um and my current thinking on the
language models is and and large models is they're not ready we don't understand them well enough yet um and you know in terms of analysis tools and and guard rails
what they can and can't do and so on to deploy them at scale because i think you know there are big still ethical questions like should an ai system always announce
that it is an ai system to begin with probably yes um it what what do you do about answering those philosophical questions about the feelings uh people may have about
ai systems perhaps incorrectly attributed so i think there's a whole bunch of research that needs to be done first um to responsibly before you know you can
responsibly deploy these systems at scale that would be at least be my current position over time i'm very confident we'll have those tools like interpretability
questions um and uh analysis questions uh and then with the ethical quandary you know i think there it's important to uh look beyond just science that's why i think
philosophy social sciences even theology other things like that come into it where um what you know arts and humanities what what does it mean to be human and the
spirit of being human and and to enhance that and and the human condition right and allow us to experience things we could never experience before and improve the the
overall human condition and humanity overall you know get radical abundance solve many scientific problems solve disease so this is the era i think this is the
amazing era i think we're heading into if we do it right um but we've got to be careful we've already seen with things like\n\nmore fundamental explanations of
physics would be the beginning you know more careful specification of that taking you walking us through by the hand as to what one would do to maybe prove those
things out maybe giving you glimpses of what things you totally missed in the physics of today exactly just here here's glimpses of no like there's a much uh a much
more elaborate world or a much simpler world or something a much deeper maybe simpler explanation yes of things right than the standard model of physics which we know
doesn't work but we still keep adding to so um and and that's how i think the beginning of an explanation would look and it would start encompassing many of the
mysteries that we have wondered about for thousands of years like you know consciousness uh life and gravity all of these things yeah giving us a glimpses of
explanations for those things yeah well um damas dear one of the special human beings in this giant puzzle of ours and it's a huge honor that you would take a pause
from the bigger puzzle to solve this small puzzle of a conversation with me today it's truly an honor and a pleasure thank you thank you i really enjoyed it thanks
lex thanks for listening to this conversation with demas establish to support this podcast please check out our sponsors in the description and now let me leave you
with some words from edskar dykstra computer science is no more about computers than astronomy is about telescopes thank you for listening and hope to see you next
time\n\nand in fact i would advocate if there's a choice building systems in the first place ai systems that are not conscious to begin with uh are just tools um
until we understand them better and the capabilities better so on that topic just not as the ceo of deep mind just as a human being let me ask you about this one
particular anecdotal evidence of the google engineer who made a comment or believed that there's some aspect of a language model the lambda language model that
exhibited sentience so you said you believe there might be a responsibility to build systems that are not essential and this experience of a particular engineer i
think i'd love to get your general opinion on this kind of thing but i think it will happen more and more and more which uh not when engineers but when people out
there that don't have an engineering background start interacting with increasingly intelligent systems we anthropomorphize them they they start to have deep
impactful um interactions with us in a way that we miss them yeah when they're gone and we sure feel like they're living entities self-aware entities and maybe even
we project sentience onto them so what what's your thought about this particular uh system was is uh have you ever met a language model that's sentient no i no no
what do you make of the case of when you kind of feel that there's some elements of sentience to this system yeah so this is you know an interesting question and uh
uh obviously a very fundamental one so first thing to say is i think that none of the systems we have today i i would say even have one iota of uh semblance of
consciousness or sentience that's my personal feeling interacting with them every day so i think that's way premature to be discussing what that engineer talked about
```

```
parser = StrOutputParser()
```

```
main_chain = parallel_chain | prompt | llm | parser
```

```
main_chain.invoke('Can you summarize the video')
```

```
'This transcript features a conversation between Lex Fridman and Demis Hassabis, CEO and co-founder of DeepMind.\n\nThe discussion begins with a playful exchange about
whether Lex himself is an AI program designed by Demis, leading into a meta-Turing test scenario.\n\nThe conversation then shifts to DeepMind\'s achievements, specific
ally **AlphaFold 2**, which solved the protein folding problem. Demis explains:\n*   **Proteins** are essential "workhorses of biology" and "bio-nano machines" whose 3
D structure dictates their function.\n*   The **protein folding problem** is predicting this 3D structure solely from a protein\'s 1D amino acid sequence.\n*   Underst
anding this 3D structure is crucial for comprehending biological functions, developing drugs, and treating diseases.\n*   This was a "grand challenge in biology" for o
```